Open problems:
Unphysicality of Hilbert spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked *research* are open research questions, not needed to check the brief. No open problem blocks verification of the brief.

From the brief

1. **Further problems.** *Research.* The list of ways in which QSTO:H fails the base conditions is open-ended: "If the reader is aware of other problems that should be added, do contact us." (Brief, Sec. 1, footnote 1.)

2. **A physically adequate model.** *Research.* The brief concludes that "the solution is not to find workarounds for these issues, but to fix the mathematical model to better reflect the physical requirements". A replacement for QSTO:H that satisfies [B]COBS, [B]WDQ and [B]STDIST is not given here; one candidate is the brief "Physical vector spaces for quantum mechanics". That candidate meets [B]WDQ only for changes of frame that map its space onto itself. For its Schwartz example, the change of frame of Proposition 3 is not one: the Gaussian ψ is a Schwartz function and $U\psi$ is not (`QuantumVectorSpaces.FurtherResults.not_forall_frameUnitary_mem_range`). A replacement therefore has to say which changes of frame it supports. It cannot support all smooth ones: no nonzero state has finite position variance in every smooth frame both at time 0 and after a free evolution (`FurtherResults.not_forall_frames_freeEvolution`; see `further-results.tex`, Sec. 2). Carcassi, Calderón and Aidala exclude the change of frame of Proposition 3 for Schwartz spaces (arXiv:2308.06669v3, Appendix, Sec. VIII). The remark after Theorem 5 that "Schwartz spaces present the same problem" concerns [B]STDIST only. A space that does not have that problem is the Gelfand–Shilov space $S^{1/2}_{1/2}(\mathbb{R}^n)$: for different n these spaces are not isomorphic as locally convex spaces (`briefs/QuantumVectorSpaces/further-results.tex`, Sec. 5, not formalized). It is not a quantum vector space in the sense of Definition 1 of that brief. (Brief, Sec. 6; Theorem 5, proof.)